Technique-Guided Coaching Feedback Generation

(a) Offline Checkpoint Design

Technique Description

Parting the Wild Horse's Mane: first left bow stance

Left bow stance; right leg naturally extended.
Turn the torso slightly to the left.
Left hand at eye level; right hand beside the right hip.

Technique excerpt shown; full text is supplied to the LLM.

Codex

Select measurable posture checkpoints

Fixed at inference

Selected Checkpoints

1 Left knee angle
2 Right knee angle
3 Left elbow angle
4 Right elbow angle
5 Normalized stance length
6 Left wrist height relative to head
7 Left wrist reach relative to head
8 Right wrist-to-hip distance

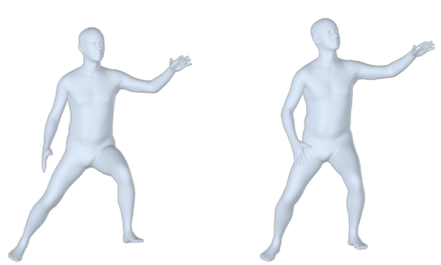
Checkpoint definitions + full technique

(b) Evidence-Based Coaching

Localized SMPL Poses

Parting the Wild Horse's Mane

Reference anchor | Student 10



22.867 s | frame0 686 | 20.788 s | frame0 623

Anchor: BV1WE411W7JB

Teacher ranges use 10 teachers.
Only the reference anchor is pictured.

Measurement & Structured Evidence

Native SMPL-24 | wrist offsets normalized by torso length

Checkpoint	Teacher ranges A / B	Student
L knee angle	[69.0, 97.4] [77.7, 99.4]	96.6°
R knee angle	[130.1, 158.4] [133.9, 152.0]	135.1°
L elbow angle	[95.5, 123.6] [98.2, 117.1]	122.5°
R elbow angle	[126.4, 133.2] [119.2, 134.0]	126.3°
Stance length / leg length	[0.502, 0.815] [0.528, 0.785]	0.422
L wrist height vs. head	[-0.188, -0.022] [-0.160, 0.009]	-0.071
L wrist reach vs. head	[0.221, 0.380] [0.196, 0.444]	0.166
R wrist-to-hip / torso	[0.558, 0.963] [0.629, 0.950]	0.484

A: median ± 2 MAD B: P10-P90 (10 teachers)

Student value: median within +/-0.2 s of the representative frame.

JSON also contains full technique, definitions, center values, window ranges and valid-frame counts.

LLM Reasoning

Text + numeric evidence only

1 Compare every checkpoint

Use both teacher reference ranges.

2 Integrate related evidence

Relate posture to the technique.

3 Assess uncertainty

Consider variability and proxy limits.

Outside range ≠ confirmed error.
Insufficient evidence → abstain.

No video or SMPL image is sent to the LLM.

Coaching Feedback

Illustration: end-pose feedback reused

Bow stance

Increase step length and deepen the front-knee bend.

Right arm

Let the arm settle naturally beside the hip; avoid excessive elbow flexion.

Left-hand height

Evidence is insufficient; no height correction is suggested.

Coach summary

Focus on a fuller bow stance and a naturally extended right arm.

Draft example: first-event poses and measurements; feedback borrowed from the third bow stance. First-event Qwen run pending.